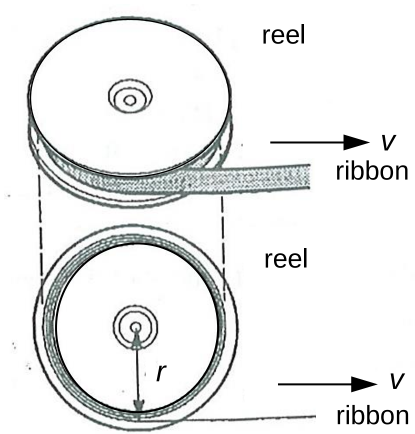
8	A ribbon, wound around a reel, is pulled and rotates about its centre. The ribbon leaves the reel at a constant v and at a variable distance r from the centre of the reel as shown.  What is the relationship between the angular velocity of the reel and the variable distance r from the centre of the reel?

	A	Angular velocity is proportional to $\frac{1}{r^2}$.
	B	Angular velocity is proportional to $\frac{1}{r}$.
	C	Angular velocity is proportional to *r.*
	D	Angular velocity does not depend on *r*.
	Answer: B $v = r\omega \quad \Rightarrow \quad \omega = \frac{v}{r}$ where ω is the angular velocity. Since *v* is constant, angular velocity ω is proportional to $\frac{1}{r}$.	

9	An object of mass *m* and weight *W* is fixed to one end of a light rod which rotates in a vertical